

Ch #14:

Fuzzy sets and fuzzy Logic

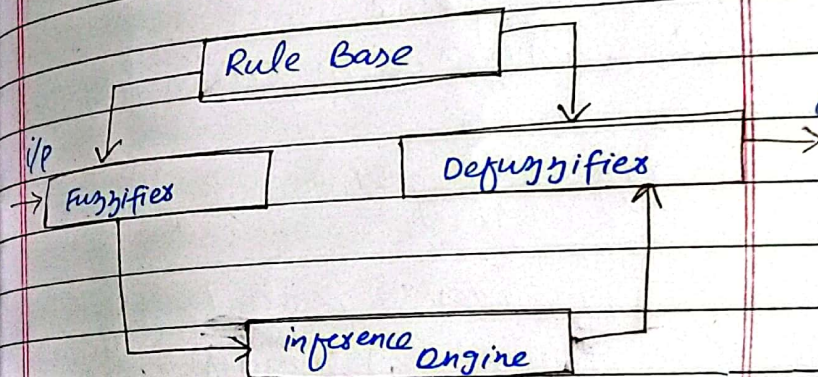
Q Analyze the characteristics of fuzzy set and fuzzy operations?

fuzzy logic:

- ⇒ Refers to a method of reasoning that deals with approximate values rather than fixed and exact values.
- ⇒ fuzzy logic allows for degree of truth.
- ⇒ mathematical method for representing uncertainty in DM.

is it cold? $\left\{ \begin{array}{l} \rightarrow \text{Yes/1} \\ \rightarrow \text{No/0} \end{array} \right\}$ boolean logic

is it cold? $\left\{ \begin{array}{l} \rightarrow \text{Yes/very/much/1} \\ \rightarrow \text{Little/0.25} \\ \rightarrow \text{very less/0.1} \end{array} \right\}$ fuzzy log

Fuzzy logic architecturefuzzy set:

• concept of fuzzy logic, where elements have degrees of membership rather than binary status.

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Fuzzy operations:

o Fuzzy Union

$$\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$$

o Fuzzy intersection

$$\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$$

o Fuzzy complement

$$\mu_{\neg A}(x) = 1 - \mu_A(x)$$

o Fuzzy difference:

$$\mu_{A-B}(x) = \min(\mu_A(x), 1 - \mu_B(x))$$

o Fuzzy aggregation

o Fuzzy cartesian product

$$\mu_{R}(x, y) = \min(\mu_A(x), \mu_B(y))$$

Applications:

- o Control systems
- o Decision making

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o List data mining tools?

- o Weka
- o CBA
- o YALE

combiner association rule mining and classification
to build classifiers based on association rules
Assoc. Rule generation → Rule sorting → Rule selection → classification

Weka:

- o Waikato environment for knowledge Analysis
- o Open source software suit for ML and DM.

Provide collection of tools for data preprocessing, regression, classification etc.

- o support scripting in java, python, & other languages

Yale:

- o Yet another learning environment
- o Open source platform
- o Also known as Rapidminer
- o Visual workflow
- o Suitable for predictive analysis